Real-world Applications and Conclusion with GitHub Copilot

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Overview

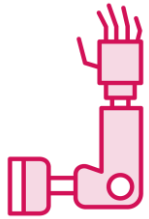
How GitHub Copilot Can Benefit IT Pros

Showcasing Practical Examples and Use Cases for GitHub Copilot

Next Steps with GitHub Copilot

How GitHub Copilot Can Benefit IT Pros

GitHub Copilot Benefits for IT Pros (Sys Admins, DevOps, SRE's Plat Engineers etc.)



The obvious:

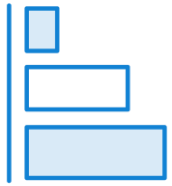
- Reduced Coding Time
- Improved Code Quality
- Lower barrier to IaC coding for beginners

Multi-lingual support for common IaC languages i.e.:

- Terraform
- Python
- PowerShell
- Crossplane

Avoid context switching stay in VS Code

- i.e. don't need to check TF registry or docs for info



Use Copilot to suggest

- variable names & descriptions to increasing quality & maintain consistency in your IaC code

Reduction of syntactical and logical errors

GitHub Copilot Next has many new things coming that IT pros (Sys Admins, DevOps, SRE's Plat Engineers etc.) will find useful

Showcasing Practical Examples and Use Cases for GitHub Copilot

Practical Examples and Use Cases for GitHub Copilot

Prototyping

Developers can use GitHub Copilot to quickly write code and prototype projects.

Copilot can generate entire functions or classes based on comments and descriptions provided by the developer.

This is especially useful for quickly testing out ideas or creating proof-of-concept implementations.

Assist non-native English speakers

GitHub Copilot understands non-English languages.

Majority of programming languages are based on American English.

Copilot can translate words from English to other languages.

Copilot can translate words like “importer” in Spanish to “import” or “colour” in British-English to “Color” in frameworks like CSS

Practical Examples and Use Cases for GitHub Copilot

Refactoring & Optimization

Copilot can assist developers in refactoring existing codebases.

By providing suggestions for more efficient or readable code, Copilot can help developers optimize their code.

It can identify patterns and propose refactoring solutions, saving time and ensuring code quality in software projects.

Learning and Education

Copilot can be used as a learning tool for programming beginners.

It can help students and learners understand coding concepts by providing contextually relevant code examples.

As learners write comments or descriptions of what they want to achieve, Copilot can offer suggestions, helping them comprehend programming s effectively.

Next Steps with GitHub Copilot

Next Steps with GitHub Copilot

Books:

- **P-AI-R Programming: How AI tools like GitHub Copilot and ChatGPT Can Radically Transform Your Development Workflow**
  - https://www.amazon.com/P-AI-R-Programming-Radically-Transform-Development/dp/B0BYRBY6V4
- **Use Github Copilot for Prompt Engineering**
  - https://www.oreilly.com/library/view/use-github-copilot/9781098162313

- Generative AI for IT Pros Path on Pluralsight:
  https://www.pluralsight.com/paths/generative-ai-for-it-pros

- User Groups: https://www.meetup.com/github

- Go work on some real-world projects in the 100DaysOfCloud community. https://www.100daysofcloud.com

- Be sure to use Pluralsight's cloud Playground (a live practice environment) & GitHub Copilot to assist you with these projects.

Summary

In this module we covered:

- Looked at the benefits of GitHub CoPilot for IT pros
- Practical GitHub Copilot examples/use cases
- Explored some next steps for furthering your journey with GitHub Copilot as an IT pro (Sys Admins, DevOps, SRE's Plat Engineers etc.)